2026 front-wing geometry study.

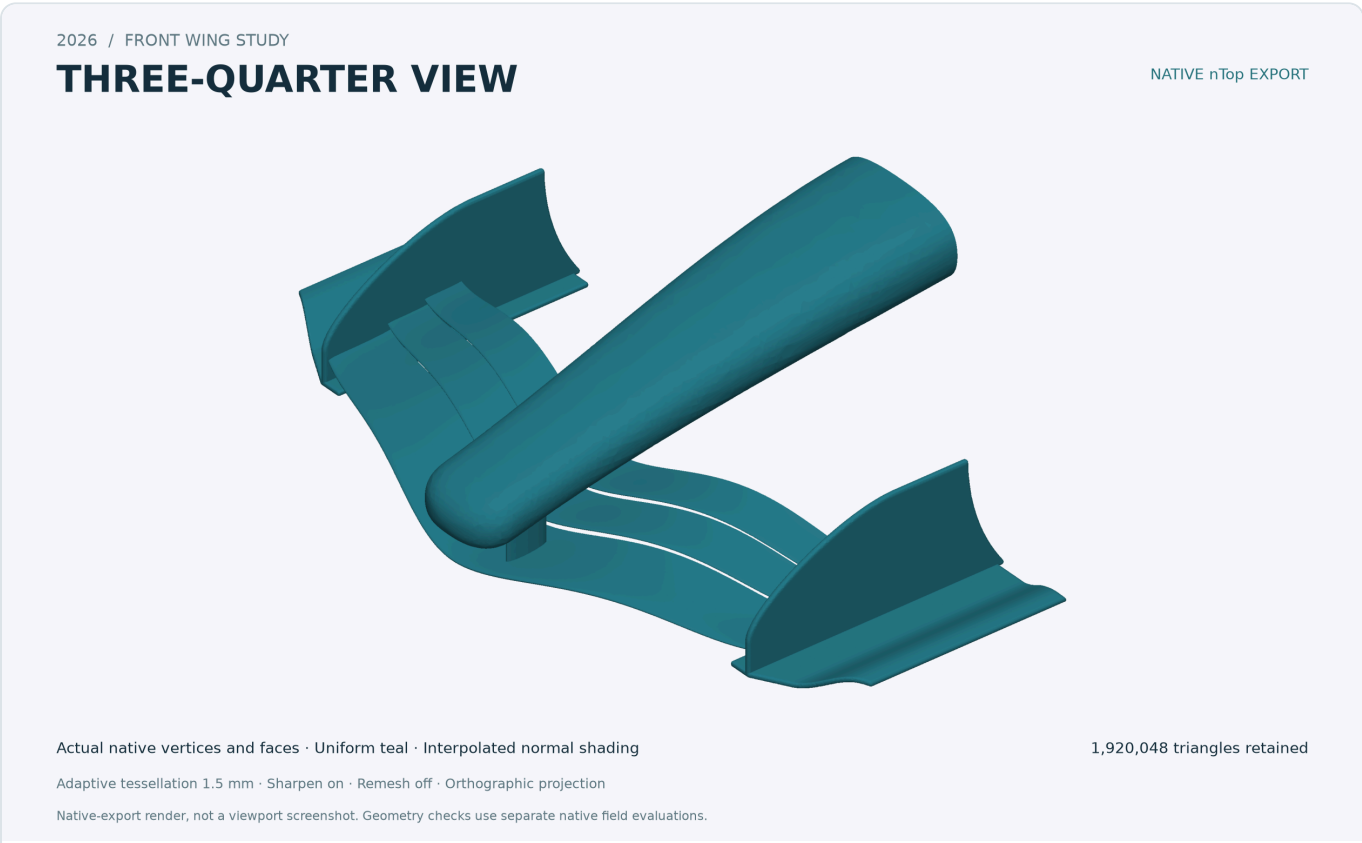
A shallow mainplane, stronger upper-element shoulders, swept endplates, rolled feet and a flatter conic nose.

> **Provisional geometric design — not FIA certification.** Reference imagery guides proportions and character. This is not a measured replica or an aerodynamically optimized design. Only evidence for this saved study applies.

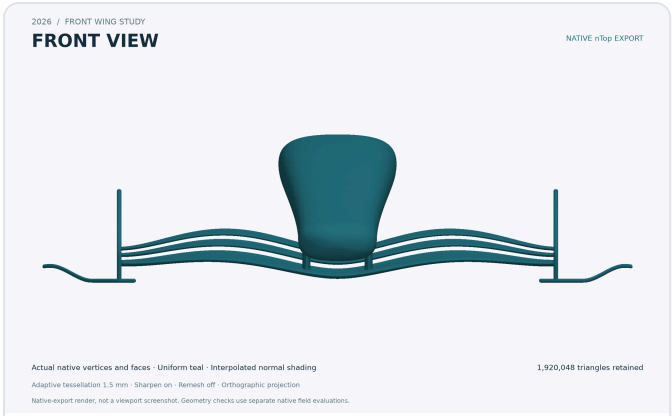
Open the editable nTop notebook

Baseline: FIA 2026 Section C, Issue 21, published and approved 19 October 2026; task date 22 October 2026. The separately published 2027 Issue 04 is not applied.

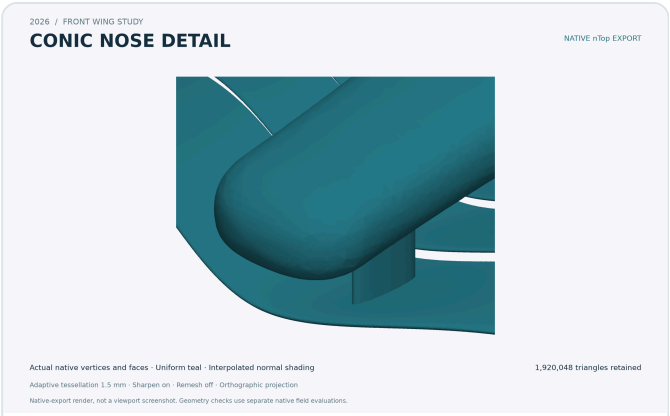
The actual model



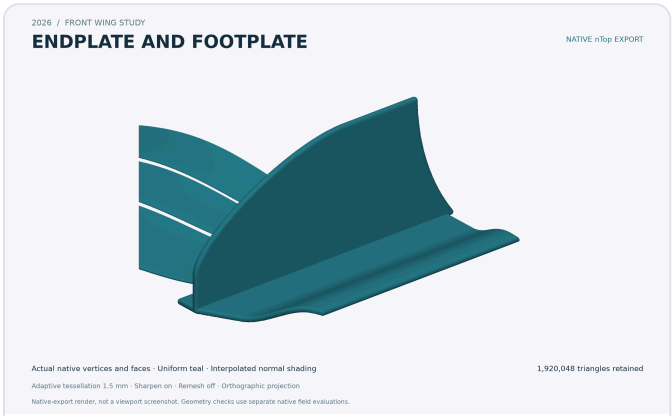
Three-Quarter View; actual native nTop AT 1.5 mm export, smooth normal shading, unchanged geometry. Not a viewport screenshot. Native-export geometry render; not a viewport screenshot.



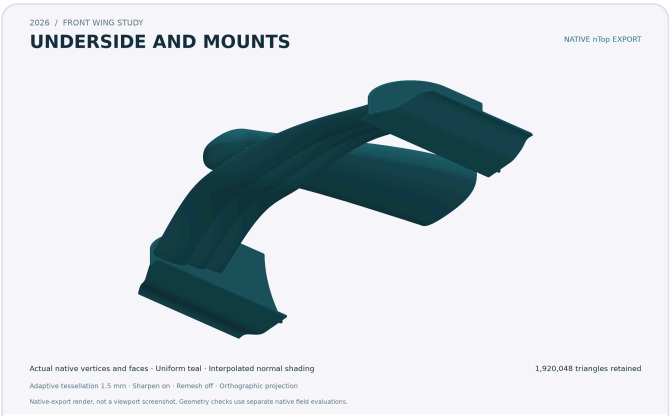
Front View; actual native nTop AT 1.5 mm export, smooth normal shading, unchanged geometry. Not a viewport screenshot. Native-export geometry render; not a viewport screenshot.



Conic Nose Detail; actual native nTop AT 1.5 mm export, smooth normal shading, unchanged geometry. Not a viewport screenshot. Native-export geometry render; not a viewport screenshot.



Endplate And Footplate; actual native nTop AT 1.5 mm export, smooth normal shading, unchanged geometry. Not a viewport screenshot. Native-export geometry render; not a viewport screenshot.



Underside And Mounts; actual native nTop AT 1.5 mm export, smooth normal shading, unchanged geometry. Not a viewport screenshot. Native-export geometry render; not a viewport screenshot.

Direct viewport capture remains unavailable. Any native-export renders above are explicitly identified; they are geometry illustrations, not measurements.

▼ Image export provenance and topology

VISUALIZATION EXPORT	RECORDED VALUE
Triangles	1920048
Connected components	1
Boundary / nonmanifold edges	0 / 5
Degenerate triangles	3
Watertight	False

The exported mesh is a discretization for illustration. Its topology does not prove exact implicit-solid validity or regulatory compliance. No manufacturing or CFD readiness is claimed.

Evidence: study_mesh_validation.json

Four nonmanifold edges and the three degenerate triangles cluster at the left mainplane/endplate junction, near XF −890.5, Y −659, Z 132.6 mm. The fifth edge is approximately 0.01 mm long on the middle-element skin near XF −787.1, Y −581.5, Z 174.0 mm. These export artifacts are retained and disclosed; the mesh is not certified watertight.

Evidence: study_degenerate_triangle_locations.json

Evidence: study_export_source_equivalence.json

Design intent and construction

Official team front and overhead launch imagery was visually inspected for its restrained nose tip, flatter crown, shallow central mainplane and stronger upper-element shoulders. These are qualitative observations, not extracted dimensions. Reference launch imagery (external branding link omitted).

1. **Shared regulatory datums.** Millimetres; X=XF increases aft, Y is spanwise and Z is upward. Nose context assumes XA=XF+100 and XC=XF−1830. These are declared vehicle assumptions.
2. **Profiles and guide rails.** Closed rounded sections feed native two-rail sweeps. Degree-four Bézier rails carry a 25 mm plan bow and a 0.06 midspan rotation coefficient. Both terms vanish at root and tip; the common map preserves section topology while raising aft elements more strongly.
3. **Endplates and feet.** One tangent cubic forms each swept endplate shoulder; its 14 mm default thickness and 6 mm finishing offset retain useful intermediate geometry. Each footplate uses two transverse cubics: 45 mm rise to Y = 875, then 1.5 mm return by Y = 895. A named 2 mm core plus 5 mm outward offset gives 12 mm nominal flat thickness. Tangent overbuild avoids coincident clipping caps.
4. **Conic nose.** Four native Conic by Points quadrants, rho 0.55, form a rounded-square section. Top/side boundary rails drive its two-rail sweep. The center guide remains an inspectable datum for the derived cap rails. The supplied CB_Conic_Nose_Cap custom block receives matching derived top/bottom/side rails and closes the front; continuity 0 requests G1. Its finite-difference tangents, regularized tip and narrow upper/lower transition band remain provisional for continuous smoothness and convexity; finite native witnesses do not certify them.
5. **Pylon burial and assembly.** A retained mainplane lower curve shifted 1 mm inward in the root-section frame is swept on the same mainplane rails. Pylon blanks are intersected with this above-skin volume, so the trim follows changing wing shape. Native underside/exclusion and overlap witnesses test the result; a union alone would not remove protrusions.

$$t=|Y|/675;\ f=1.5t^2-0.5t^3;\ b=16t^2(1-t)^2$$

$$X=X_0+\mathrm{sweep}\cdot f+25b+s\cdot q_x-w\cdot q_z$$

$$Z=Z_0+\mathrm{rise}\cdot f+A\cdot b+w\cdot q_x+s\cdot q_z$$

$$w=\mathrm{tip_rotation_coefficient}\cdot f+0.06b$$

The actual section angle is atan2(w,s). Distances scale by $\sqrt{(s^2+w^2)}$, so the common map preserves their ratios, not their absolute values. Internal coefficients are design choices, not FIA limits.

Twelve primary inputs

Use the notebook INPUTS group. Change a working copy, inspect the visible guards and geometry, then restore defaults to reproduce this report. Nose construction, fixed shaping coefficients, regulatory values and derived placements are named internally. Individual input ranges do not guarantee valid coupled configurations.

INPUT	UNITS	DEFAULT	EFFECT	SUPPORTED STUDY RANGE / COUPLING
Root leading-edge height	mm	95	Raises all three profiles and the pylon root. Coupled with tip rise and section camber.	92 to 100
Outboard rise	mm	40	Smooth spanwise rise; zero derivative at the symmetry plane.	38 to 43
Slot lift	mm	40	Element 2 height offset; element 3 is derived as 1.75 times this value. This is NOT the spherical slot gap.	39 to 41
Trailing-edge rise ratio	dimensionless	0.12	Camber/incidence control: trailing height divided by each root chord. Both skins share the endpoint.	0.115 to 0.125
Rear element root chord	mm	120	Changes third element chord; tip assembly chord remains 333 mm through derived taper.	118 to 125
Leading-edge sweep	mm	225	Root-to-tip X displacement of the common leading datum. Cubic easing preserves centerline tangency.	223 to 227
Root trailing-edge radius	mm	0.7	Native Fillet Curves radius before the two-rail sweep; section scaling changes the outboard radius.	0.65 to 0.8
Endplate thickness	mm	14	Total lateral endplate thickness, including6mm rounded edge offset per side. Core thickness is derived as input minus12mm.	14 to 18
Pylon lateral station	mm	95	Positive-side pylon center; negative side is mirrored. Front impact structure interface only.	90 to 105

INPUT	UNITS	DEFAULT	EFFECT	SUPPORTED STUDY RANGE / COUPLING
Pylon chord	mm	100	Streamlined pylon profile chord; lateral thickness is fixed at 24 mm maximum.	90 to 110
Spanwise shoulder height	mm	35	Adds a smooth midspan elevation shoulder; zero at center and tip.	25 to 45
Tip section twist	deg	-3	Rotates complete sections progressively across span; negative lowers the trailing edge.	-4 to 0

Evidence: study_controls.json

Independent geometric checks

Wing and coupled controls

WING CHECK	RESULT	VALUE / CRITERION	SCOPE
Continuous profile RV containment	PASS	3.8368 mm conservative margin	Bernstein support certificate in floating arithmetic; analytical profiles and material-removing edge fillets only. Native correspondence is separate.
Coupled parameter study	PASS	38 cases	Finite combinations; no guarantee for every interior control combination.
Study-wide RV certificate	PASS	1.3309 mm worst conservative margin	Continuous geometry for each listed case, not the entire input domain.
Worst outboard trailing thickness	9.9415 mm	Limit 10 mm	Sampled last-40-mm thickness criterion; small remaining margin requires attention.
Worst normal angle / trailing plan radius	20.556° / 343.17 mm	Limits 30° / 200 mm	Analytical derivatives sampled over the surfaces; normal-rule interpretation remains provisional.
Default 3D slot samples	8.8028–12.408 mm	Required spherical gap 5–15 mm	Sampled distances are upper bounds on the continuous minimum.
Sensitive-case 3D slot samples	7.4561–10.377 mm	Required spherical gap 5–15 mm	Finite neighboring-span search; not a continuous spherical-gauge proof.

Evidence: study_validation.json · study_parameter_studies.json

Nose and mounting footprint

These margins describe the independent conic/cap equations. Native correspondence must be established for both the swept body and the supplied cap. Requesting G1 in the cap block does not certify exact tangent continuity through its finite-difference and field-transition implementation.

NOSE CHECK	VALUE	ANALYTICAL STATUS	SCOPE
RV roof	7.4031 mm	PASS	Finite samples of the independent cap/body description.
RV floor	15.781 mm	PASS	Finite samples of the independent cap/body description.
RV plan	13.632 mm	PASS	Finite samples of the independent cap/body description.
RS plan coverage	17.424 mm	PASS	Finite samples of the independent cap/body description.
Section radius: forward / XA0	42.987 / 61.346 mm	PASS	Limits 20 / 45 mm; rational derivative stationary roots, including endpoints.
Pylon mounting-footprint margin	3.0298 mm	PASS	Finite footprint boundaries over 38 cases and pylon extrema.
Default vertical wing clearance	34.992 mm	PASS	Vertical separation, not minimum Euclidean distance.

Evidence: study_nose_spec.json · study_nose_analytical.json

▼ Nose construction limitations

Envelope and full-footprint checks are finite samples, not continuous certificates. Cap equations transcribed independently from supplied source; native witnesses required before correspondence is established. G1 uses finite-difference rail derivatives; exact G2 is not claimed. Effective cap heights differ above/below side level. Tiny-tip regularization excluded from exact-tip claims. Vehicle coordinates assumed; physical tests, crash structure and complete-car interfaces remain unverified. Convexity concerns reconstructed rational quadrants. Native upper/lower height selection has a narrow transition band UpperWeight=clamp(0.5+UpRatio*100000,0,1); continuous convexity through that band and exact tip remains provisional.

Endplate and footplate

ENDPLATE / FOOTPLATE CRITERION	RESULT	SCOPE
Endplate RV / required side silhouette	PASS / PASS	Exact cubic support extrema and polygon/circle bounds under the intended offset construction.
Footplate required top silhouette	PASS	Unchanged convex footprint contains the required polygon.

ENDPLATE / FOOTPLATE CRITERION	RESULT	SCOPE
Footplate flat nominal thickness	12 mm	Side-visible normal thickness must be ≥10 mm; nominal flat thickness does not establish all junctions.
Footplate curve 1 radius after ideal offset	52.87 mm	Finite transverse samples; concave limit50 mm. Euclidean offset equivalence is provisional at mixed trim/skin junctions.
Footplate curve 2 radius after ideal offset	79.398 mm	Finite transverse samples; concave limit50 mm. Euclidean offset equivalence is provisional at mixed trim/skin junctions.

Evidence: study_endplate_validation.json · study_footplate_validation.json

Pylon underside burial

CONSTRUCTION BOUND	VALUE	SCOPE
Mapped complete pylon region: root X	40.075–162.47 mm	Inside the 0–200 mm swept trim profile; conservative algebraic bounds over stated controls.
Minimum skin separation after 1 mm burial	5.048 mm	Root-section bound; conditional on native sweep/trim correspondence.

The interval enclosure covers the declared input domain rather than only sampled footprints. It establishes the intended burial inside the mainplane; native underside and overlap checks test implementation correspondence.

Evidence: sources/study_pylon_trim_bounds.json

Global mixed trim/skin offsets, finished junction curvature and complete normal thickness remain provisional. Earlier work demonstrated that a composed native field offset need not equal an ideal Euclidean dilation. Transverse radius estimates alone do not certify every edge.

Fresh native validation

CLI evaluation and geometric acceptance are separate. Surface residuals are native field values; strict sign witnesses establish local inside/outside relationships.

NATIVE CASE	CLI	GEOMETRY / GUARD	EVIDENCE
default	PASS	PASS	736 witnesses; max surface residual 0.00095896 mm
default guards	PASS	PASS	True, True, True, True, True, True, True
invalid shoulder 55	PASS	Expected rejection	False, False, True, True, True, True, False
low envelope	PASS	PASS	326 witnesses; max surface residual 0.00032244 mm
high envelope	PASS	PASS	326 witnesses; max surface residual 0.00031719 mm

NATIVE CASE	CLI	GEOMETRY / GUARD	EVIDENCE
wide gap	PASS	PASS	326 witnesses; max surface residual 0.0003069 mm

▼ **Default component and underside checks**

GROUP	RESULT	WITNESSES
wing and junctions	PASS	51
pylon underside	PASS	120
pylon wing overlap	PASS	120
endplate footplate	PASS	134
nose pylon overlap	PASS	30
nose	PASS	281

Evidence: study_native_validation.json

Evidence: study_export_source_equivalence.json

▼ **Native validation limits**

Native witnesses establish finite surface/sign correspondence, not continuous topology/regulatory proof. Underside witnesses cover15 sites per pylon. Separate new-law algebraic domain bounds support the intended shared-rail trim across the complete stated input ranges; they remain conditional on native correspondence. Unprovided component manifests are explicitly omitted; no previous nose/footplate evidence is silently substituted. Conic body witnesses do not certify the custom rounded cap unless a separate cap manifest is supplied.

Restored defaults and reopening

SAVED-STATE CHECK	RESULT
Notebook reopened	PASS
All twelve defaults restored	PASS
Important native outputs	PASS
Default guards	PASS

Evidence: study_reopen_validation.json

Regulation and remaining boundaries

Fixed-wing scope: no driver-adjustable mechanism is fitted. This notebook does not establish complete 2026 race architecture, structural loads, deflection, laminate qualification, crash approval or homologation. CFD was not run.

C3.10.9 requires pylons to remain hidden from below and removes material below the profiles. C3.7.1 permits a flatter convex nose crown if its radius, tangency, envelope and silhouette conditions hold; C3.12.4 restricts nose/wing intersections. The 20/45 mm nose section-radius checks and cap behavior are distinct from impact-structure approval.

▼ All 27 baseline rule groups — authority, requirements and checks

CLAUSE	REQUIREMENT	ENFORCEMENT / CHECK	CURRENT SCOPE
C2.1; C2.2	Use millimetres; X increases rearward, Y right, Z up. Y=0 symmetry, Z=0 sprung-car datum, XF=0 front axle at Legality Setup.	Use XF as notebook X origin. Shared datum planes; explicitly mark vehicle mounting datum provisional.	Current-study geometry evidence above; remaining continuous checks provisional.
C2.3.1; C2.4	General width |Y| <=950 mm; component-specific reference volumes remain controlling.	Check body extrema and exact component RV containment; 950 mm alone is insufficient.	Current-study geometry evidence above; remaining continuous checks provisional.
C3.2.2; C3.18.1	Rigid, continuous, solid, impervious aerodynamic surfaces; moving-body exceptions are specifically limited. Linear elastic behaviour and further FIA tests apply.	Geometry checks cannot establish rigidity or physical surface properties.	Physical testing / complete-car context required.
C3.2.3	Nominal symmetry about Y=0 and tangent continuity across it. Limited manual flap-adjustment asymmetry exception.	Mirror shared geometry and use zero spanwise derivative at center. Do not use asymmetry exception for ECU motion.	Current-study geometry evidence above; remaining continuous checks provisional.
C3.2.4–C3.2.6	Individual components generally single simply-connected volumes and one section per axis plane unless exempt. Trimming only overlapping material; permitted unions tangent-continuous. Prescribed fillets are true tangent circular arcs.	Retain pre-trim components; check connectivity and section counts. Generic smooth-union is not proof of permitted Fillet Radius.	Current-study geometry evidence above; remaining continuous checks provisional.
C3.3.1–C3.3.3	Digital CAD compliance; intentional edge-of-limit designs need the stated justification. Physical ±3 mm manufacturing and ±2 mm FW positioning allowances are not CAD-envelope allowances.	Use design margin. No silent enlargement of RV. Full-car datum/inspection is outside this notebook.	Physical testing / complete-car context required.
C3.10.1(a–d); App C2 §21–22; §34.10	Up to three nonintersecting simply-connected profile volumes and up to three Y sections; profiles inside RV-FW-PROFILES and cover RS-FW-PROFILES in plan.	Construct exact RV polygon/extrusion/trims; compare each complete body with RV; plan-projection coverage and section count checks.	Current-study geometry evidence above; remaining continuous checks provisional.
C3.10.1(e–f)	Every section trailing extremum visible below; all but the last section trailing extremum hidden above.	Require positive chord overlap and correct vertical stacking. Check visibility in each sampled Y plane; samples are not continuous proof.	Current-study geometry evidence above; remaining continuous checks provisional.

CLAUSE	REQUIREMENT	ENFORCEMENT / CHECK	CURRENT SCOPE
C3.10.1(g)	Within final 40 mm of each section: lower-visible tangent slopes positive; upper-to-lower distance at most 10 mm beyond Y400, 15 mm inside Y400.	Use thin rising trailing sections; validate derivatives and distance, including Y400 boundary. Treat generic vertical-thickness estimate as provisional.	Current-study geometry evidence above; remaining continuous checks provisional.
C3.10.1(h)	No concave curvature below. Upper-visible concave curvature radius at least 50 mm.	Select convex lower profile; evaluate signed curvature on both sides. Check local blends as well as nominal sections.	Current-study geometry evidence above; remaining continuous checks provisional.
C3.10.1(i–j)	Surface normals within 30 degrees of a vertical plane normal to RS-FW-SECTION; exception for surfaces exactly at Y0 or Y675 and conditional pre-edge-radius assessment.	Sample surface normals against the specified plane; angle interpretation and exhaustive proof remain provisional.	Current-study geometry evidence above; remaining continuous checks provisional.
C3.10.1(k)	At every span location, spherical-gauge minimum distance between adjacent profile volumes 5–15 mm.	Check true 3D shortest separations, not just vertical gap or profile-point spacing. Reject coupled changes failing the sampled clearance check.	Current-study geometry evidence above; remaining continuous checks provisional.
C3.10.1(l)	Each section rear extremum projected to Z0 traces one tangent-continuous curve with radius at least 200 mm.	Use smooth trailing rails, zero center derivative; check analytic curve curvature if available. Tessellated fit alone provisional.	Current-study geometry evidence above; remaining continuous checks provisional.
C3.10.1 concluding paragraph	Optional rear-element upper trailing-edge Gurney at most 10 mm high with listed limited exemptions.	Omit Gurneys for baseline; if added, enforce RV and remaining rules.	Current-study geometry evidence above; remaining continuous checks provisional.
C3.10.2; App C2 §23	Endplate body within its RV; up to two Y sections; convex radius at least 5 mm and concave at least 100 mm.	Build curved body with tire-cylinder and sloping-top envelope. Validate before combinations.	Current-study geometry evidence above; remaining continuous checks provisional.
C3.10.3–C3.10.5; App C2 §24–26	Outer and inner footplates have distinct RVs. Outer footplate radius >=5 mm and specified concave radii >=50 mm; up to two sections in specified X/Z regions. Optional diveplane has 300×325×50 mm box plus its RV and curvature limits.	Model genuine joined footplates covering plan RS; do not replace with a narrow upright. Omit optional diveplane.	Current-study geometry evidence above; remaining continuous checks provisional.
C3.10.6; App C2 §34.11–12	Combined endplate includes body and both footplates; discard body below inner footplate first. Single simply-connected volume; side-visible thickness >=10 mm; full top and side reference-surface coverage. Fillets <=10 mm except diveplane junction <=40 mm.	Check thickness, silhouette coverage, connectivity and permitted fillets separately. A simple plate fails broad top coverage.	Current-study geometry evidence above; remaining continuous checks provisional.

CLAUSE	REQUIREMENT	ENFORCEMENT / CHECK	CURRENT SCOPE
C3.10.7; App C2 §33.9	Pylon within XF[-1200,-950], Y[50,150], Z[60,275]; any Z section area <=6000 mm² and Y thickness <=35 mm.	Use symmetric streamlined mounts, check each independent section and below-view obscuration. Real attachment design requires vehicle context.	Physical testing / complete-car context required.
C3.10.8	Optional strake confined to its RV and 775×20×100 mm box, with <=2 Y/Z sections.	Omit baseline strake; optional does not mean unconstrained.	Current-study geometry evidence above; remaining continuous checks provisional.
C3.10.9; C3.10.12	Assembly includes profiles/endplates/pylons, optionally strake. Discard profile parts outboard of endplate and pylon below profiles. Allowed junction fillets <=10 mm. Single volume and no overlaps; pylon hidden below.	Retain meaningful pre-trim components and verify final connectivity, tangent junctions and hidden mount silhouette.	Current-study geometry evidence above; remaining continuous checks provisional.
C3.10.10(a–j)	Primary moving part excludes forwardmost profile; secondary is only rearmost profile. Pivot points inside flap and within25 mm of local leading extremum; inner points Y<150, outer Y475–675. Rotation boundaries cover full chord, normals within30° of Y.	If movement fitted, build named axis and rotation surfaces with fixed support relation; inspect points and joint boundaries.	No adjustable mechanism fitted; fixed-wing study.
C3.10.10(k–r)	Tool-only stationary manual adjustment may not lower specified incidence; point displacement <=30 mm primary/60 mm secondary. ECU adjustment about one axis only; two fixed modes, Straight reduces incidence; transition <=400 ms; <=2 actuators and linked sensors, symmetric.	Separate design incidence from actuation state. Validate both states and full motion envelope; timing, ECU and sensors need hardware verification.	No adjustable mechanism fitted; fixed-wing study.
C3.10.10(s–w)	Moving flaps stay inside profile RV, retain required internal relationships, obey remaining bodywork rules with explicit exemptions. Physical stops prevent RV exit, failure returns Corner Mode; deployment only stationary or activation zone.	Show state limits and swept clearance; sampled angle sweep provisional. Do not assert fail-safe or operational compliance from rotation animation.	No adjustable mechanism fitted; fixed-wing study.
C3.10.11	Auxiliary brackets, hangers, fishplates and fairings each constrained. <=8 hangers, >=150 mm spacing for same adjacent pair; <=3 rotation brackets per flap; hangers plus brackets <=10. Most auxiliaries hidden below.	Omit unnecessary auxiliaries. If modeled record counts, bounds, below-view visibility, attachment and fillet limits.	Current-study geometry evidence above; remaining continuous checks provisional.
C3.18.2	1000 N downward each side at XF=-925,Y=±525,Z=300 using prescribed adapter; allowable deflection15 mm symmetric/20 mm single-sided.	Physical load/deflection test required relative to survival-cell datum. No structural certification in geometry notebook.	Physical testing / complete-car context required.

CLAUSE	REQUIREMENT	ENFORCEMENT / CHECK	CURRENT SCOPE
C3.18.3–C3.18.4	Flap trailing edge <=10 mm deflection under60 N normal in Corner Mode. Endplate <=7 mm under60 N vertical or lateral; prescribed25 mm pad.	Physical tests required; no inference from valid solid geometry.	Physical testing / complete-car context required.
C3.18.15; C3.19.1	FIA monitoring markers/cameras and DOC-050 procedures may apply. Endplate uses PL-ANTISPLINTER; fasteners/inserts >=30 mm behind endplate LE.	Manufacture, laminate qualification, fastener and inspection checks outside present geometry. No access to nonpublic DOC-050 details assumed.	Physical testing / complete-car context required.

Any continuous claims above apply only to the documented mathematical construction and listed cases. Spherical slot gauges, unresolved surface-normal interpretation, global junctions and full parameter-domain validity remain provisional where not separately established.

A targeted official-source search found no additional public clarification for the normal-angle or spherical-gap interpretation. Issue 21 remains the authority. Clarification-search record.

Files and sources

- Current editable front-wing study notebook
- Official FIA 2026 Section C, Issue 21 · Saved official PDF
- Official FIA technical-regulation index · Full rule register
- Reference morphology and regulatory review

The nose custom block is an embedded snapshot of the user-supplied dependency. Its source identity is recorded in the nose specification. The supplied dependency is retained as CB_Conic_Nose_Cap.ntop; keep this folder with the notebook when distributing the project.

Package provenance

Packaging note: evidence text labels and file references were neutralized for this distribution. Retained provenance hashes identify the original pre-label-cleanup source files, not the renamed or sanitized package copies. Historical command paths describe the recorded validation runs; those temporary notebooks, scripts and failed runs are not distributed. The packaged notebook is linked to the earlier geometry by package_geometry_equivalence.json. Technical results, numeric values and recorded source hashes are preserved.

Packaged notebook SHA-256: `d9d60720adebd6bde670e6a2a95997c58ff6587cecd493b2931cd56ec0a67288`.

Strict annotation-only geometry equivalence · Packaged notebook fresh native validation · Actual native illustration mesh · Image and evidence packaging checks.

Prepared from the packaged study evidence. Earlier design results are not substituted for missing current-study checks.